



Comprehensive analysis of widely targeted metabolic profiling and transcriptome reveals the grafting-mediated stress tolerance

Junxing Guo^{a,c}, Jieqiang He^{a,c}, Abid Khan^b, Xuewei Li^{a,c}, Chundong Niu^{a,c}, Qingmei Guan^{a,c,*}

^a State Key Laboratory for Crop Stress Resistance and High-Efficiency Production, Yangling, Shaanxi 712100, China

^b Department of Horticulture, The University of Haripur, Haripur 22620, Pakistan

^c Shaanxi Key Laboratory of Apple, College of Horticulture, Northwest A&F University, Yangling, Shaanxi 712100, China

ARTICLE INFO

Keywords:

Apple
Graft
Stress tolerance
Transcriptome
Metabolomics

ABSTRACT

Grafting is widely used in fruit trees for propagation and improvement of the scion's stress tolerance and fruit quality. *Malus prunifolia* 'Fupingqiu' ('FPQZ') is a unique germplasm with strong environmental stress tolerance and can enhance scion tolerance when used as a rootstock. Previous studies have demonstrated altered DNA methylation patterns and enhanced expression of stress-related genes in scion tissue when 'Golden Delicious' (*Malus × domestica*) was grafted onto *M. prunifolia* 'FPQZ'. However, the global metabolic changes involved in the grafting events remain unclear. In this study, we investigated metabolite and gene differences between *M. prunifolia* 'FPQZ' and 'Golden Delicious' using the widely targeted UPLC-MS/MS-based metabolic approach, and further analyzed the effects of grafting on the scion and rootstock. Metabolomic analysis suggested that *M. prunifolia* 'FPQZ' accumulated higher levels of metabolites, including phenylpropanoids, flavonoids, and lipids, compared with 'Golden Delicious'. After grafted onto the *M. prunifolia* 'FPQZ', 154 differentially accumulated metabolites were identified in 'Golden Delicious', including a large proportion of flavonoids, suggesting a critical role of flavonoid metabolism in grafting-mediated stress tolerance. Furthermore, exogenous application of naringenin and dihydroquercetin enhanced drought tolerance in apple plants, highlighting their functional roles in grafting-mediated stress responses. Overall, our study provides a comprehensive analysis and new insights into grafting-induced metabolic reprogramming and transcriptional regulation.

Key message: Our study offers a comprehensive analysis of grafting effects on metabolic changes and transcriptional regulation, also highlights the critical roles of flavonoid metabolites in conferring stress tolerance through grafting.

Introduction

Grafting is an essential asexual reproduction technique in apple production. During grafting, the scion and the rootstock are in contact through artificial wounds, and the graft union is formed through three stages: initial adhesion (Pina and Errea, 2005), callus formation via dedifferentiation at the interface (Flaishman et al., 2008), and the establishment of new vascular bundle bridge (Moore, 1982; Wang and Kollmann, 1996). Following graft healing, the re-establishment of the

vascular bundles at the graft union provides the basis for scion and rootstock interaction (Sidorova et al., 2021), including the exchange of water, nutrients, inorganic substances (Tsipouridis and Thomidis, 2005; North and Cook, 2008), and metabolites (Aloni et al., 2010). Previous studies have shown that defense-related metabolites such as caffeic acid and shikimic acid accumulate in scions of heterografted grapevines, whereas 2-hydroxy-pyridine accumulates in the rootstock (Tedesco et al., 2021). Similarly, a large number of metabolites were discriminated in grafted tea and non-grafted tea, and most of these metabolites are involved in phenylpropanoids and flavonoid biosynthetic pathways, such as quercetin 3-galactoside, theasinensin B, quercetin 3-O-glucoside, and astragalin (Qi et al., 2019).

Compared with other asexual reproduction methods such as cuttings and layering, grafting offers unique advantages, including preserving desirable traits of cultivars while conferring additional adaptive characteristics (Warschefsky et al., 2016). Grafting can influence the

Peer review under Responsibility of the Editorial Board of Journal of Integrative Horticulture.

* Corresponding author at: State Key Laboratory for Crop Stress Resistance and High-Efficiency Production, Yangling, Shaanxi 712100, China.

E-mail address: qguan@nwfau.edu.cn (Q. Guan).

<https://doi.org/10.1016/j.jih.2026.100004>

Received 7 January 2026; Received in revised form 13 April 2026; Accepted 19 May 2026

3051-2905/© 2026 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

phenotypes of both scion and rootstock. Typically, scion is regarded as the recipient and is grafted onto different rootstocks to acquire specific traits such as dwarfism (Koepke and Dhingra, 2013), resistance to biotic and abiotic stresses (Venema et al., 2008; Shahid et al., 2019), and improved fruit quality and yield (Tsipouridis and Thomidis, 2005; North and Cook, 2008; Turhan et al., 2012). However, emerging evidence suggests that scions can also influence rootstock traits. For example, in tuber crops, grafted scions can significantly enhance crop yields. (Lardizabal and Thompson, 1990), although such studies in fruit trees remain limited.

In addition to its effects on physiological traits and substances communication, grafting also contributes to gene expression, epigenetic modification, and other molecular regulation (Dong et al., 2022). DNA methylation variation caused by heterograft has been reported in *Malus* (Li et al., 2022), *Citrus* (Huang et al., 2021), and *Cucurbitaceae* (Avramidou et al., 2015) plants, with the extent and pattern of methylation changes varying among species. Grafting has also been shown to affect gene expression in both homografted and heterografted plants. Transcriptome analyses have identified numerous differentially expressed genes (DEGs) associated with grafting in apple (Li et al., 2022), cucurbit (Liu et al., 2020b), watermelon (Wang et al., 2020), and grapevines (Yang et al., 2015). Notably, some of these DEGs are involved in stress responses and are upregulated in heterografted scions, including homologs of *HSPs* and *CBFs* in 'Golden Delicious' grafted onto *M. prunifolia* 'FPQZ' (Li et al., 2022). Recent studies suggested that some macromolecules, including RNAs (Haywood et al., 2005; Thieme et al., 2015; Martin et al., 2009; Pant et al., 2008) and proteins (Liu et al., 2020a), may also act as signaling molecules that transferring between scion and rootstock and play a regulatory role.

M. prunifolia (FPQZ) is a native wild apple species in China. Although its small and acidic fruits limit its commercial value (Li et al., 2021), it exhibits strong tolerance to various environmental stresses, including drought, cold, heat, and salt stress, as well as heavy metal toxicity, making it a widely used rootstock (Wang et al., 2012; Huang et al., 2018). 'HanFu' (*M. domestica*) apple grafted onto *M. prunifolia* rootstock shows enhanced tolerance to copper toxicity through alterations in Cu transport, antioxidant defense mechanisms, and gene expression patterns (Wan et al., 2021). Similarly, 'Golden Delicious' (*M. domestica*) exhibits improved tolerance to drought, heat, and cold stress when grafted onto *M. prunifolia* 'FPQZ' (Li et al., 2022). Further analyses have revealed significant upregulation of genes associated with molecular chaperones, cuticular wax biosynthesis, abscisic acid (ABA) biosynthesis, and lipoxygenase pathways in grafted 'Golden Delicious', including homologs of *HSP*, *KCS19*, *LOX*, and *NCED3* (Li et al., 2022). However, the metabolites involved in the graft-mediated stress tolerance of 'Golden Delicious' onto *M. prunifolia* are still unelucidated.

Metabolomics is employed to describe the dynamics of endogenous small molecule metabolites in specific organisms or tissues. Transcriptomics, on the other hand, provides insights into global biological processes in gene expression (Chen et al., 2014; Yang et al., 2020; Zheng et al., 2019; Li et al., 2020). The integration of metabolomics and transcriptomics facilitates the elucidation of complex biological mechanisms. The multi-omics approach has been widely used in horticultural plants, such as tomatoes (Osorio et al., 2020), apples (Xu et al., 2020), citrus (Tietel et al., 2011), and peppers (Sun et al., 2021). For example, Osorio et al. combined the metabolomics and targeted transcriptome profiling techniques to investigate the changes in amino acids and multiple secondary metabolites during fruit ripening in tomatoes, revealing the molecular mechanisms underlying tomato fruit ripening and overripening (Osorio et al., 2020). In apple, similar approaches have elucidated metabolic and molecular regulation of apple fruit during the development and ripening (Xu et al., 2020). However, less attention has been paid to how grafting affects the metabolic and molecular regulation of scion and rootstock.

In the present study, we first performed grafting events using the cultivated apple 'Golden Delicious' grafted onto the wild apple *M.*

prunifolia 'FPQZ'. By integrating a widely targeted UPLC-MS/MS-based metabolic approach with transcriptome analysis, we compared the differential metabolites and genes between *M. prunifolia* 'FPQZ' and 'Golden Delicious', as well as the effect of grafting event on both the 'Golden Delicious' scion and *M. prunifolia* 'FPQZ' rootstock.

Materials and methods

Plant materials and growth conditions

Three-year-old 'Golden Delicious' and *M. prunifolia* 'FPQZ' trees were used as the plant materials. Branch buds from *M. domestica* 'Golden Delicious' were harvested in the winter and grafted onto *M. prunifolia* 'FPQZ' in the following spring. Mature leaves from one-year-old branches were collected in summer and quickly frozen in liquid nitrogen for subsequent metabolic profiling and transcriptome analysis. There are three biological replicates, and every three plants were collected as a replicate. Two-month-old 'GL-3' plants were application with 0.2 mM naringenin and 0.2 mM dihydroquercetin for 7 days, then subjected to short-term drought stress. The drought stress treatment and peroxidase (POD) activity, catalase (CAT) activity, and hydrogen peroxide (H_2O_2) content were measured as previously described, with minor modifications (Chen et al., 2020). All plants were planted in plastic pots (12 cm × 13 cm) filled with a mixture of soil and substrate (PINDSTRUP, Denmark) (1:1, v/v) and grown in the experimental station at Northwest A&F University, Yangling (34°20'N, 108°24'E), Shaanxi Province, China.

Metabolite extraction and UPLC-MS sample preparation

The liquid nitrogen-frozen leaves were ground into fine powder using a mixer mill (MM 400, Retsch) with a zirconia bead for 1.5 min at 30 Hz. Approximately 100 mg powder was weighed and extracted overnight at 4°C with 1.0 ml 70% aqueous methanol. Following centrifugation at 10,000 × g for 10 min, the extracts were absorbed (CNWBOND Carbon-GCB SPE Cartridge, 250 mg, 3 ml; ANPEL, Shanghai, China) and filtrated (SCAA-104, 0.22 μm pore size; ANPEL, Shanghai, China) before LC-MS analysis.

Metabolite detection by UPLC-QTRAP-MS/MS

The HPLC and mass spectrometry conditions were given below:

HPLC conditions: The sample extracts were analyzed using an LC-ESI-MS/MS system (HPLC, Shim-pack UFLC SHIMADZU CBM30A system; MS, Applied Biosystems 6500 Q TRAP). The analytical conditions were as follows, 2 μl sample extracts were injected into a Waters ACQUITY UPLC HSS T3 C18 column (1.8 μm, 2.1 mm × 100 mm) working at 40°C and a flow rate of 0.4 ml/min; solvent system of A (water with 0.04% formic) and B (acetonitrile with 0.04% formic) was used for elution; the gradient program was as follow: 95:5 (v/v) at 0 min, 5:95 (v/v) for 11 min, 5:95 (v/v) for 12 min, 95:5 (v/v) for 12.1 min, 95:5 (v/v) for 15 min. The effluent was alternatively connected to an ESI-triple quadrupole-linear ion trap (Q TRAP)-MS.

ESI-Q TRAP-MS/MS: LIT and triple quadrupole (QQQ) scans were acquired on a triple quadrupole-linear ion trap mass spectrometer (Q TRAP), API 6500 Q TRAP LC/MS/MS System, equipped with an ESI Turbo Ion-Spray interface, operating in a positive ion mode and controlled by Analyst 1.6.3 software (AB Sciex). The ESI source operation parameters were as follows: ion source, turbo spray; source temperature 500°C; ion spray voltage (IS) 5500 V; ion source gas I (GSI), gas II (GSII), curtain gas (CUR) was set at 55, 60, and 25.0 psi, respectively; the collision gas (CAD) was high. Instrument tuning and mass calibration were performed with 10 and 100 μmol/L polypropylene glycol solutions in QQQ and LIT modes, respectively. QQQ scans were acquired as MRM experiments with collision gas (nitrogen) set to 5 psi. DP and CE for individual MRM transitions was done with further DP and CE

optimization. A specific set of MRM transitions were monitored for each period according to the metabolites eluted within this period. Analyst 1.6 software was applied for data processing, and for the integration of peak areas the IntelliQuan algorithm was used.

Metabolomics data analysis

To ensure data reliability, equal aliquots of all samples were pooled to generate quality control (QC) samples and submitted for metabolomics analysis. For Cluster analysis (CA), the metabolite data were standardized and then clustered using pheatmap v1.0.12. Principal component analysis (PCA) was conducted using log₂-transformed data with the SIMCA-P 14.0 software. Differentially accumulated metabolites (DAMs) were identified based on a variable importance in projection (VIP) value threshold ≥ 1 and a fold change ≥ 2 or fold change ≤ 0.5 . KEGG pathway enrichment analysis of DAM was performed using the clusterProfiler and ggplot2 R package (Kanehisa and Goto, 2000).

RNA-seq analysis

The RNA-seq and data analysis methods were consistent with those reported in the previous study (Li et al., 2022). Briefly, for non-grafted plants and grafted plants, leaves from the scions and rootstocks were collected and used for RNA extraction. RNAs were subjected to sequencing on Illumina HiSeq 4000 platform by Novogene (Beijing, China). Three biological replicates were performed for each experiment. After sequencing, clean reads were mapped to the latest *Malus × domestica* genome (GDDH13 v1.1, https://iris.angers.inra.fr/gddh13/downloads/GDDH13_1-1_formatted.fasta.bz2) using HISAT2 v2.1.0. Reads counting within genes were obtained using HTSeq v0.12.4 tool and matching gff3 format annotation file (https://iris.angers.inra.fr/gddh13/downloads/gene_models_20170612.gff3.bz2). Differentially expressed genes were produced by DESeq2 v1.30.1 R package with thresholds of adjusted p-value below 0.05 and $\log_2$ (fold change) greater than 1. Gene Ontology analyses were performed using agriGO v2.0 and clusterProfiler v3.18.1.

Results

Distinct metabolomic profiles distinguish grafted and non-grafted apple tissues

To reveal the metabolites involved in *M. prunifolia* 'FPQZ' mediated stress tolerance, we performed UPLC-MS-based metabolomic analysis on four sample groups, including 'Golden Delicious' (*M.dom*), *M. prunifolia* 'FPQZ' (*M.pru*), 'Golden Delicious' scion (*gM.dom*) grafted onto *M. prunifolia* 'FPQZ' rootstock (*gM.pru*). Principal component analysis (PCA) on these metabolites showed a distinct separation of sample metabolites into four groups as well as strong repeatability in three replicates (Fig. 1A). Cluster analysis (CA) of these metabolites also showed similar results, with heterogeneity between groups and homogeneity within groups (Fig. 1B). These results indicated that the metabolites in different sample groups were significant separation and reliable for further analysis. A total of 668 metabolites were identified across all four groups (Supplemental Table 1), which were classified into amino acids and derivatives, carbohydrates, flavonoids, lipids, nucleotides and derivatives, organic acids and derivatives, phenylpropanoids, vitamins and derivatives, and terpene.

Enhanced flavonoid and lipid accumulation of 'FPQZ' compared with 'GD'

Given that *M. prunifolia* 'FPQZ' exhibits excellent resistance to abiotic stresses such as drought, cold, heat and salt compared with most of the apple cultivars (Wang et al., 2012; Li et al., 2022), we first compared the metabolome of *M. prunifolia* 'FPQZ' and 'Golden

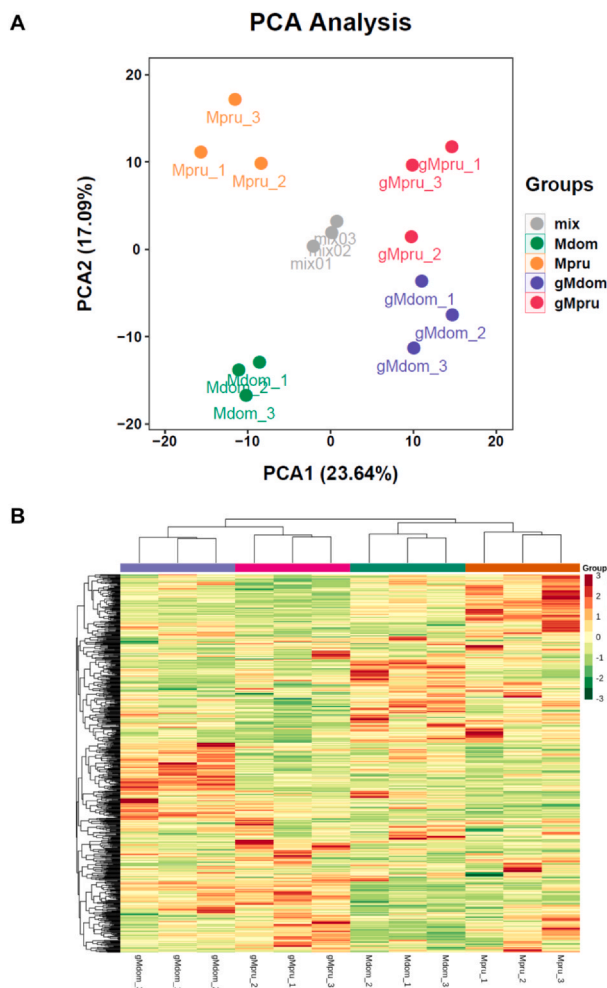


Fig. 1. Quality control of the Widely Targeted Metabolic Profiling. (A) PCA analyze of 15 metabolite datasets of the different samples. Each point represents one metabolite profiling experiment. There were three biological repeats for each group. Mdom: *Malus domestica* 'Golden Delicious'; Mpru: *Malus prunifolia* 'Fupingqiuizi'; gMdom: scion of 'Golden Delicious' grafted onto 'Fupingqiuizi'; gMpru: rootstock of 'Golden Delicious' grafted onto 'Fupingqiuizi'; mix: mixture of all 12 samples, quality control sample. (B) Cluster analysis of the metabolites detected in the total samples. Red indicates high abundance, green indicates low abundance. Metabolites were divided into four main clusters.

Delicious'. Results showed that 116 differentially accumulated metabolites (DAMs) were identified in *M. prunifolia* 'FPQZ' (63/116 accumulated more) when compared with 'Golden Delicious' (53/116 accumulated more) (Supplemental Table 2). These 116 metabolites included 10 amino acids and their derivatives, 48 phenylpropanoids and flavonoid compounds, 21 organic acids and their derivatives and 11 lipids, etc. Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analysis revealed that these differential metabolites were mostly enriched in 'flavonoid biosynthesis', 'isoflavonoid biosynthesis', 'phenylpropanoid biosynthesis', 'vitamin B6 metabolism', 'purine metabolism', 'linoleic acid metabolism', 'glutathione metabolism', 'caffeine metabolism', and 'anthocyanin biosynthesis' metabolic pathways (Figs. 2A and 2B), indicating a wide variation of metabolic capability between *M. prunifolia* 'FPQZ' and 'Golden Delicious'.

Flavonoids are known to be involved in biotic stress and abiotic stress tolerance (Nakabayashi et al., 2014; Yaginuma et al., 2002). We found that the flavonoid metabolites accounted for a large proportion in the differential metabolites between *M. prunifolia* 'FPQZ' and 'Golden Delicious', with 34 differential flavonoid compounds representing 29.5% of the total differential metabolites (Supplemental Table 2). In

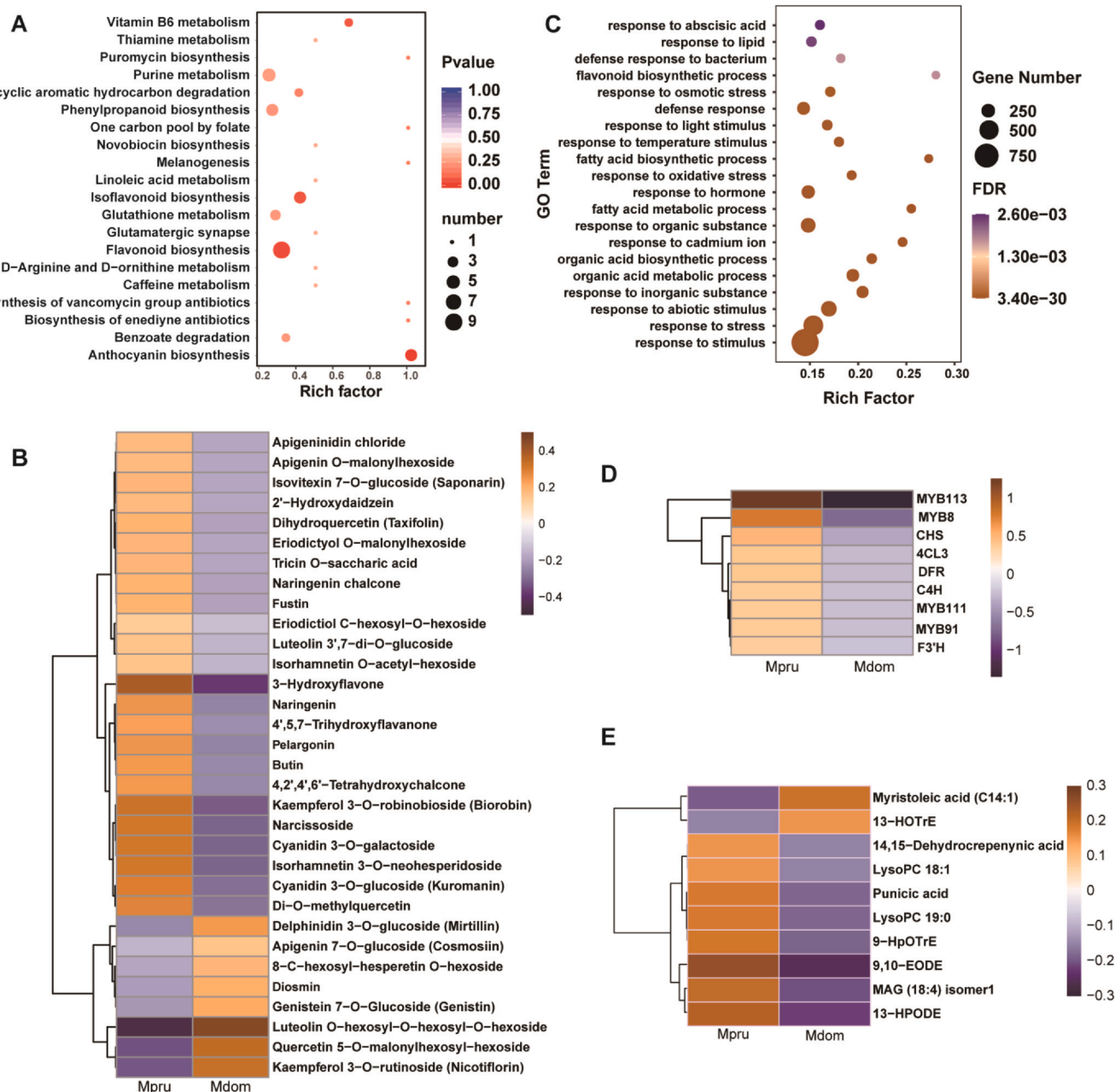


Fig. 2. Differentially detected metabolites and differential expression genes between *Malus prunifolia* and *Malus domestica*. (A) KEGG enrichment of *Malus prunifolia* and *Malus domestica* differentially accumulated metabolites. The size of the points represents the number of differentially expressed genes, and different colours represent different Q-values. The larger the rich factor is, the higher the degree of enrichment. (B) GO enrichment of differentially expressed genes between *Malus prunifolia* and *Malus domestica*. (C) Heat map of differentially accumulated flavonoids between *Malus prunifolia* and *Malus domestica*. (D) Partial flavonoids biosynthesis related genes which has higher expression level in *Malus prunifolia*. (E) Heat map of differentially accumulated lipids between *Malus prunifolia* and *Malus domestica*. Brown indicated high abundance; purple indicated low abundance.

addition, 'flavonoid biosynthesis' was found the most enriched metabolic pathway by KEGG analysis (Fig. 2A). Among the 34 differential flavonoids compounds, most of them (25 compounds) showed higher contents in *M. prunifolia* 'FPQZ' compared with 'Golden Delicious' (Fig. 2B), such as naringenin chalcone, naringenin, apigenin, dihydroquercetin and so on. Pathway mapping indicated that several of these compounds are key intermediates in flavonoid biosynthesis (Fig. 3).

Transcriptomic analysis further supported these findings. Gene Ontology (GO) enrichment analysis showed significant enrichment in flavonoid biosynthesis process, fatty acid metabolic process, response to osmotic stress, response to stress and response to lipid (Fig. 2C). More importantly, expression levels of genes encoding key flavonoid

metabolites related enzymes, including *chalcone synthase* (*CHS*), *cinnamate 4-hydroxylase* (*C4H*), *4-coumarate: coenzyme A ligase* (*4CL*), *flavanone-3-hydroxylase* (*F3'H*), *dihydroflavonol 4-reductase* (*DFR*), and *UDP-dependent glycosyltransferase* (*UGT*), showed higher transcriptional activity in *M. prunifolia* 'FPQZ' than in 'Golden Delicious' (Fig. 3, Fig. 2D). In addition, MYB transcription factors that positively regulate flavonoid biosynthesis including MYB113, MYB8, MYB111, and MYB91 also showed higher expression levels in *M. prunifolia* 'FPQZ' (Koyama et al., 2014) (Fig. 2D), consistent with higher flavonoid contents. Comparative analysis of DAMs and DEGs involved in the flavonoid biosynthesis pathway between *Malus prunifolia* 'FPQZ' and 'Golden Delicious' revealed a coordinated regulation that contributes to the elevated flavonoid levels in *M. prunifolia* 'FPQZ'. Specifically, the

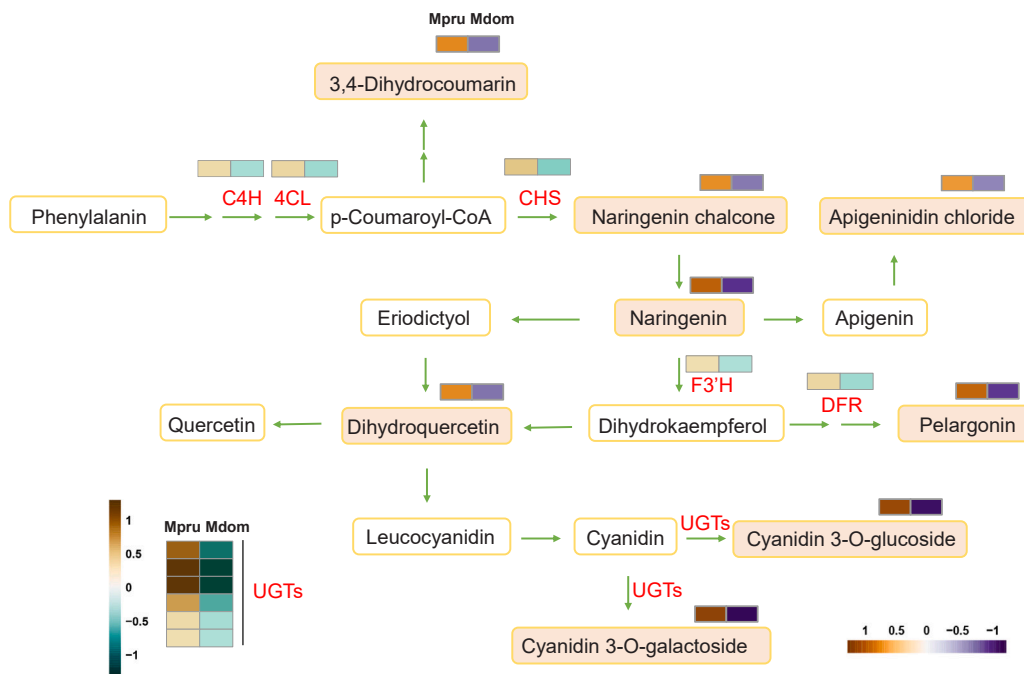


Fig. 3. Differentially regulated metabolites and genes related to the flavonoid biosynthesis pathway between *Malus prunifolia* and *Malus domestica*. The red box indicates higher metabolite abundance in *Malus prunifolia* plants, the heat maps (brown-purple) above the metabolites show differential metabolites abundance between *Malus prunifolia* and *Malus domestica*. The red words indicate higher gene expression levels in *Malus prunifolia* plants, the heat maps (brown-green) above the genes show differential gene expression level between *Malus prunifolia* and *Malus domestica*.

upregulated expression of *C4H*, *4CL*, and *CHS* in *M. prunifolia* 'FPQZ' promotes the biosynthesis and accumulation of naringenin chalcone and naringenin. In addition, the higher expression of *DFR* facilitates the accumulation of pelargonidin (Fig. 3). Collectively, these transcriptional and metabolic changes lead to enhanced flavonoid accumulation in *M. prunifolia* 'FPQZ'.

Lipids also play crucial roles in plant response to stress (Hou et al., 2016) and we focused on the difference in lipid metabolism between *M. prunifolia* 'FPQZ' and 'Golden Delicious' as well. The results showed that most differentially accumulated lipid metabolites in the two groups have higher contents in *M. prunifolia* 'FPQZ' than the 'Golden Delicious', accounting for 80% of the 10 lipid differential metabolites. The 8 higher accumulation of lipid metabolites in *M. prunifolia* 'FPQZ' were lysophosphatidylcholine (LysoPC) (18:1), MAG (18:4) isomer1, lysophosphatidylcholine (LysoPC) (19:0), Punicic acid, ($\pm$) 9(10)-epoxy-12Z-octadecenoic acid (9,10-EODE), 14,15-Dehydrocrepenynic acid, 9-HpOTrE and 13-HPODE (Fig. 2E) respectively.

Grafting induces coordinated metabolic reprogramming in both scion and rootstock

To explore the common effects of grafting on both scion and rootstock, we overlapped the differential metabolites in grafted 'Golden Delicious' (as the scion) and *M. prunifolia* 'FPQZ' (as the rootstock). A total of 54 metabolites were identified, of which 25 increased and 29 decreased after grafting (Fig. 4A, Supplemental Table 3). Among these metabolites, flavonoids accounted for 46%, followed by lipids (11%), amino acids and derivatives (9%), organic acids and derivatives (9%), carbohydrates (9%), terpenoids (6%), phenylpropanoids (4%), and others (11%) (Fig. 4B), indicating the potential roles of these metabolites in the grafting events. Interestingly, all 5 carbohydrates (ribulose-5-phosphate, D-Glucose 6-phosphate, D-Fructose 6-phosphate, glucose-1-phosphate and D-Fructose 6-phosphate-disodium salt) accumulated higher in the grafted tissues as compared with the non-grafted plants (Fig. 4C). These carbohydrates are the initial compounds of the carbohydrates metabolism in apples, and may affect the carbohydrate metabolism of the whole plants (Yoon et al., 2021; Stincone et al., 2015). Transcriptomic analysis identified 358 shared DEGs between scion and rootstock after grafting (Fig. 4D). Consistently, GO analysis suggested that these DEGs were enriched in various metabolic

processes including carbohydrate metabolic and catabolic process, lipid, organic acid and carboxylic acid metabolic processes (Fig. 4E).

After grafting, 154 different metabolites were identified in 'Golden Delicious', as compared with the non-grafted 'Golden Delicious', with 99 metabolite contents increased and 55 metabolite contents decreased (Fig. 4A, Supplemental Table 4). KEGG analysis indicated enrichment in pathways including biosynthesis of secondary metabolites, flavonoid biosynthesis, flavone and flavanol biosynthesis, isoflavonoid biosynthesis, and carbon metabolism (Fig. 5A), which was consistent with the increased tolerance of grafted 'Golden Delicious' to abiotic stresses (Li et al., 2022). GO analysis revealed that 4421 DEGs involved in response to stimulus and metabolic process were significantly enriched in 'Golden Delicious' after grafting (Fig. 5B). These data indicated that grafting-induced stress tolerance is closely associated with metabolic reprogramming, particularly in flavonoid biosynthesis. Furthermore, we mapped the DAMs and DEGs to the flavonoid biosynthesis pathway, and the results showed that the contents of quercetin and phlorizin were increased in 'Golden Delicious' after grafting. Besides, the flavonoids such as naringenin chalcone and naringenin, were also increased in both 'Golden Delicious' and *M. prunifolia* 'FPQZ' after graft event (Fig. 5C), meanwhile, *C4H*, *4CL*, and *CHS*, which encode key enzymes in the biosynthesis of naringenin chalcone and naringenin, were significantly upregulated in grafted 'Golden Delicious', thereby promoting the accumulation of these metabolites (Fig. 5D). In addition, the MYB family transcription factor genes *MYB8* and *MYB113* were also upregulated, contributing to the regulation of flavonoid biosynthesis in grafted 'Golden Delicious'. The comparative analysis indicates that, at both the transcriptional and metabolic levels, *Malus prunifolia* 'FPQZ' as a rootstock enhances flavonoid accumulation in the scion, the enhancement may further contribute to improved stress tolerance through indirect regulatory effects.

We also analyzed the effects of grafting on the rootstock *M. prunifolia* 'FPQZ'. After grafting, 128 differential metabolites were identified in *M. prunifolia* 'FPQZ' (Fig. 4A, Supplemental Table 5). KEGG analysis showed that these differential metabolites were enriched in various metabolic pathways including starch and sucrose metabolism, arginine biosynthesis, zeatin biosynthesis, etc (Supplemental Figure 1). GO analysis of the 1559 DEGs in *M. prunifolia* 'FPQZ' after grafting revealed that these DEGs are involved in response to stimulus and their metabolic processes were enriched (Supplemental Figure 2).

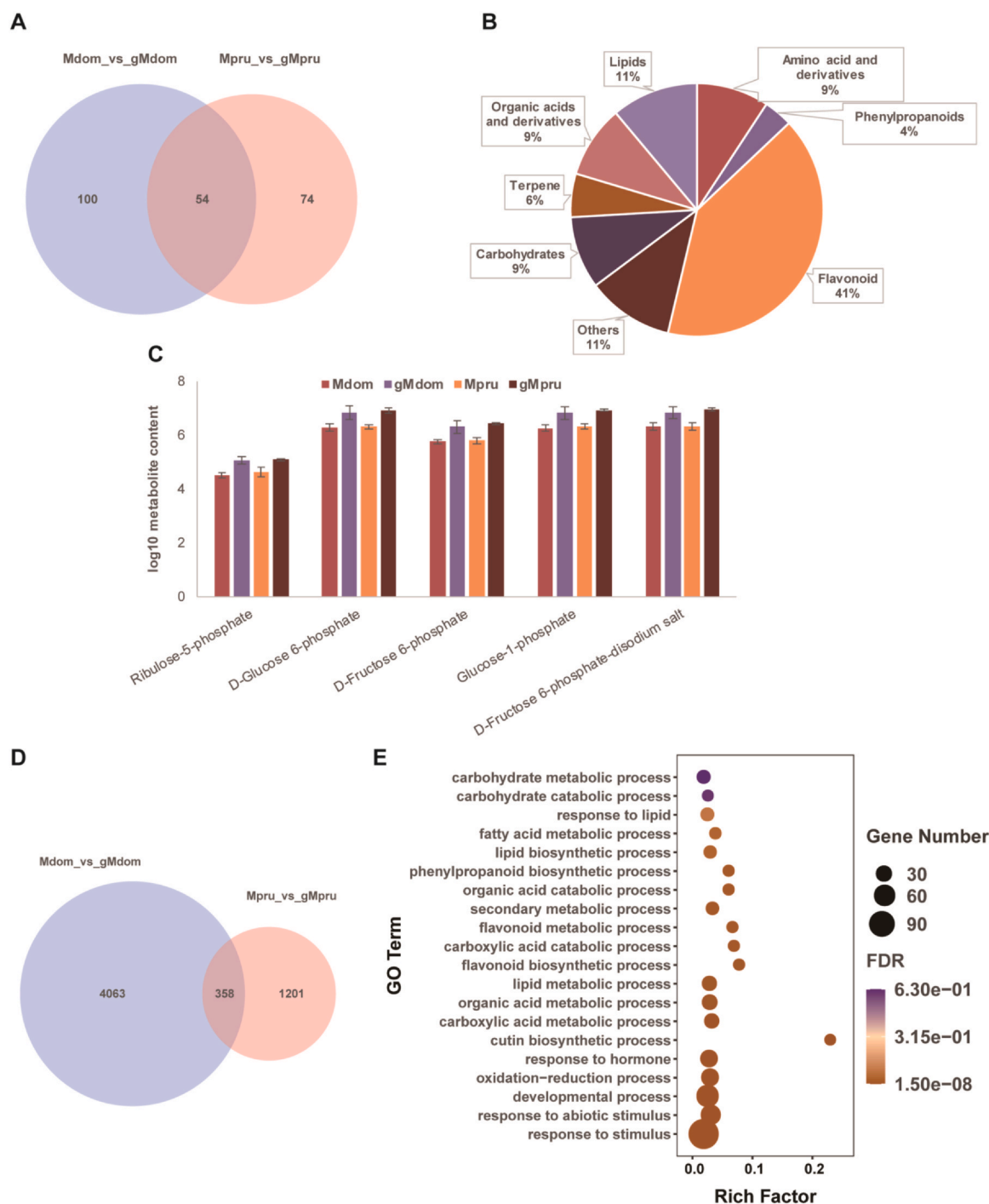


Fig. 4. Analysis of differential accumulated metabolites and expression genes in graft events. (A) Venn diagram of differential accumulated metabolites. (B) Composition of the common 54 metabolites in “Mdom_vs_gMdom” group and “Mpru_vs_gMpru” group. (C) Carbohydrates contents detected in Mdom, gMdom, Mpru and gMpru groups. (D) Venn diagram of differential expression genes. (E) GO enrichment of the 358 overlap genes displayed in (D).

Flavonoid metabolites naringenin and dihydroquercetin contribute to drought tolerance

Naringenin and dihydroquercetin are vital metabolites in plant flavonoid biosynthesis pathway and are known to participate in stress responses especially drought stress. To explore the effects of naringenin and dihydroquercetin in drought tolerance, we first irrigated two-month-old ‘GL-3’ plants using 0.2 mM naringenin or 0.2 mM dihydroquercetin for 7 days, then these plants and normal watered ‘GL-3’ plants (control) were subjected to short-term drought stress. Under drought stress, the leaves of the control plants wilted faster than those

of naringenin- and dihydroquercetin-treatment plants, indicating the treatment of naringenin and dihydroquercetin can improve the plants tolerance to drought (Fig. 6A). We also measured hydrogen peroxide (H_2O_2) content, peroxidase (POD) activity and catalase (CAT) activity to assess the ROS level and capacity of the antioxidant defense system under drought stress. After drought stress, the naringenin- and dihydroquercetin-treatment plants accumulated less H_2O_2 than control plants, indicated lower ROS levels in naringenin- and dihydroquercetin-treatment plants (Fig. 6C). Conversely, the activity of POD and CAT were significantly higher in naringenin- and dihydroquercetin-treatment plants than control plants under drought conditions (Fig. 6 B, D).

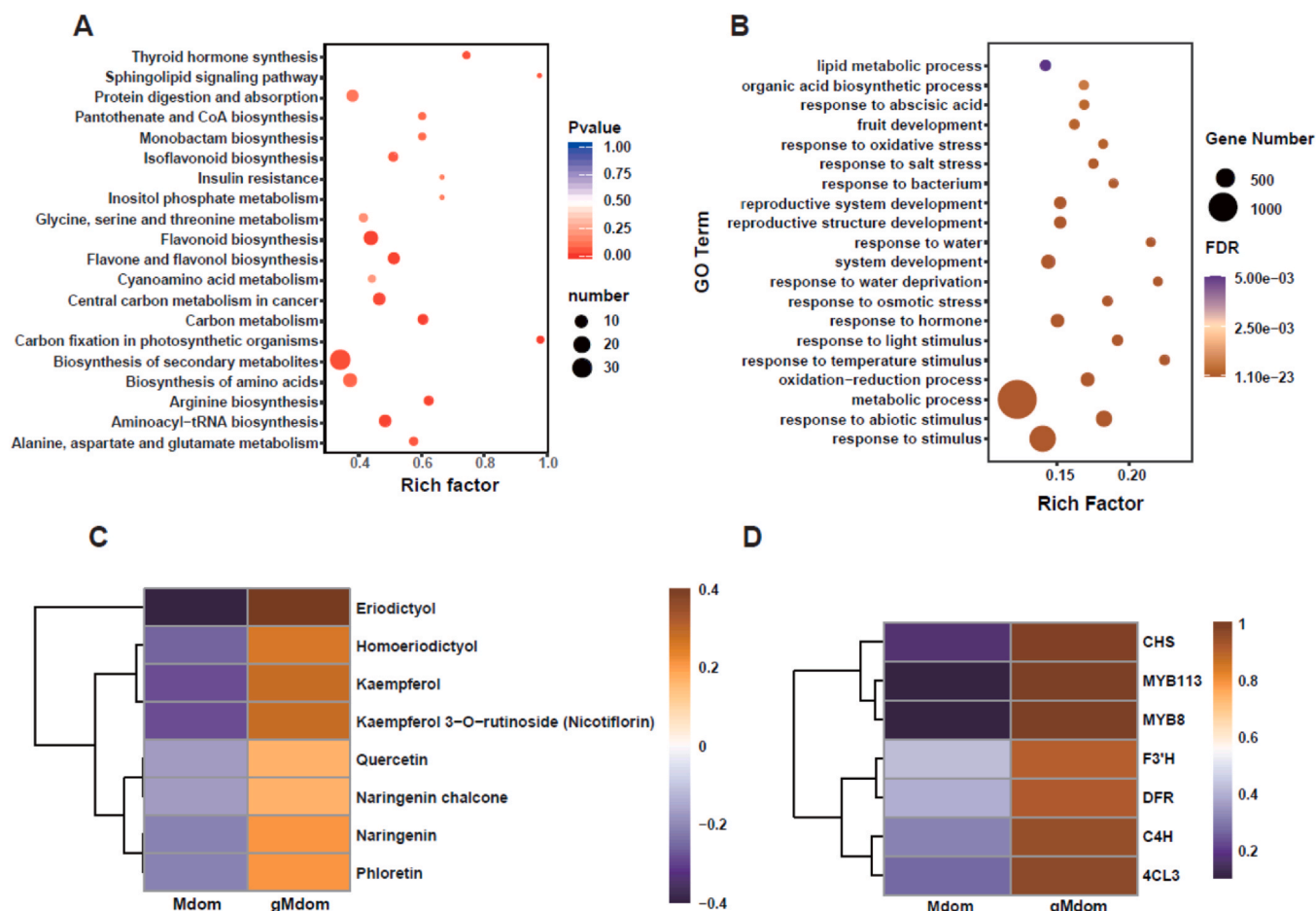


Fig. 5. Enrichment of differentially accumulated metabolites and differentially expressed genes before and after grafting of *Malus domestica*. (A) KEGG enrichment of differentially accumulated metabolites in *Mdom_vs.gMdom*. (B) GO enrichment of differentially expressed genes in *Mdom_vs.gMdom*. (C) Heat map of differentially accumulated flavonoids between *Malus domestica* and grafted-*Malus domestica*. (D) Partial flavonoids biosynthesis related genes which has higher expression level in grafted-*Malus domestica*. Brown indicated high abundance; purple indicated low abundance.

Therefore, the above results suggest that the treatment of naringenin and dihydroquercetin increases the activities of CAT and POD, then promotes H_2O_2 scavenging, thereby improving the tolerance of apple 'GL-3' plants to drought.

Discussion

M. prunifolia 'FPQZ' is known for its tolerance to harsh environmental conditions and is particularly adapted to drought, cold, heat, and disease challenges (Li et al., 2022), making it one of the best rootstocks in northwest China where extreme weather conditions often occur. *M. prunifolia* is also widely used as a breeding parent for stress tolerance; for example, Michurin once used it to improve apple hardiness (Sokolov et al., 2015). In current study, we compared the metabolome of *M. prunifolia* 'FPQZ' and 'Golden Delicious'. The results showed that phenylpropanoids, flavonoids and lipids compounds were significantly accumulated in *M. prunifolia* 'FPQZ' (Supplemental Table 2). Compounds in phenylpropanoids pathway are considered to be critical metabolites in plant stress tolerance (Geng et al., 2020; Yaginuma et al., 2002). Flavonoids can enhance the resistance to *Alternaria alternata* f. sp. *mali* in tomatoes (Wei et al., 2017) and oxidative as well as drought stress in *Arabidopsis* (Nakabayashi et al., 2014). Naringenin has been reported to protect plants from osmotic stress by regulating the expression of stress tolerance-related genes and modulating the antioxidant capacity in bean (Ozfidan-Konakci et al., 2020). Apigenin and dihydroquercetin are also distinguished antioxidants. Apigenin and dihydroquercetin pretreatment can enhance the rice

defense system under salinity stress by regulating iron uptake and translocation and activating downstream antioxidant genes (Mekawy et al., 2018; Liang et al., 2013). In this study, we also showed the close relationship between drought and flavonoids, the pretreatment of naringenin and dihydroquercetin improved the tolerance of apple 'GL-3' plants to drought stress, increased the antioxidant enzymes POD and CAT activity to scavenge H_2O_2 (Fig. 6). Lipids can also play important roles in stress tolerance (Hou et al., 2016). Lipids are important components of biological membranes and research showed that they can maintain normal cellular metabolism and response to stress as well (Okazaki and Saito, 2014). Lysophospholipids has been reported to participate in stress responses and plant-environment interactions in California poppy (Viehweiger et al., 2002). Research also suggested a direct involvement in abiotic and biotic stress responses of free fatty acids like Linoleic acid and its derivatives (9,10-EODE, 9-HpOTrE, 13-HPODE and so on) in plants (Okazaki and Saito, 2014; Upchurch, 2008). Therefore, increased accumulation of phenylpropanoids and flavonoids compounds and lipids may contribute to the stress tolerance of *M. prunifolia* 'FPQZ'.

Grafting is an ancient horticultural technique that is currently one of the most common commercial techniques for production of fruit crops, vegetable, and ornamentals (Mudge et al., 2009). For three thousand years, grafting has contributed to the improvement and development of fruit crops. It serves various purposes, including rapid propagation of scion cultivars, increasing stress tolerance (Gilardi et al., 2013; Reeves et al., 2022); improving fruit quality or even production (Al-Harbi et al., 2018; Ghanem et al., 2011; Ikinci et al., 2014; Shi et al., 2019);

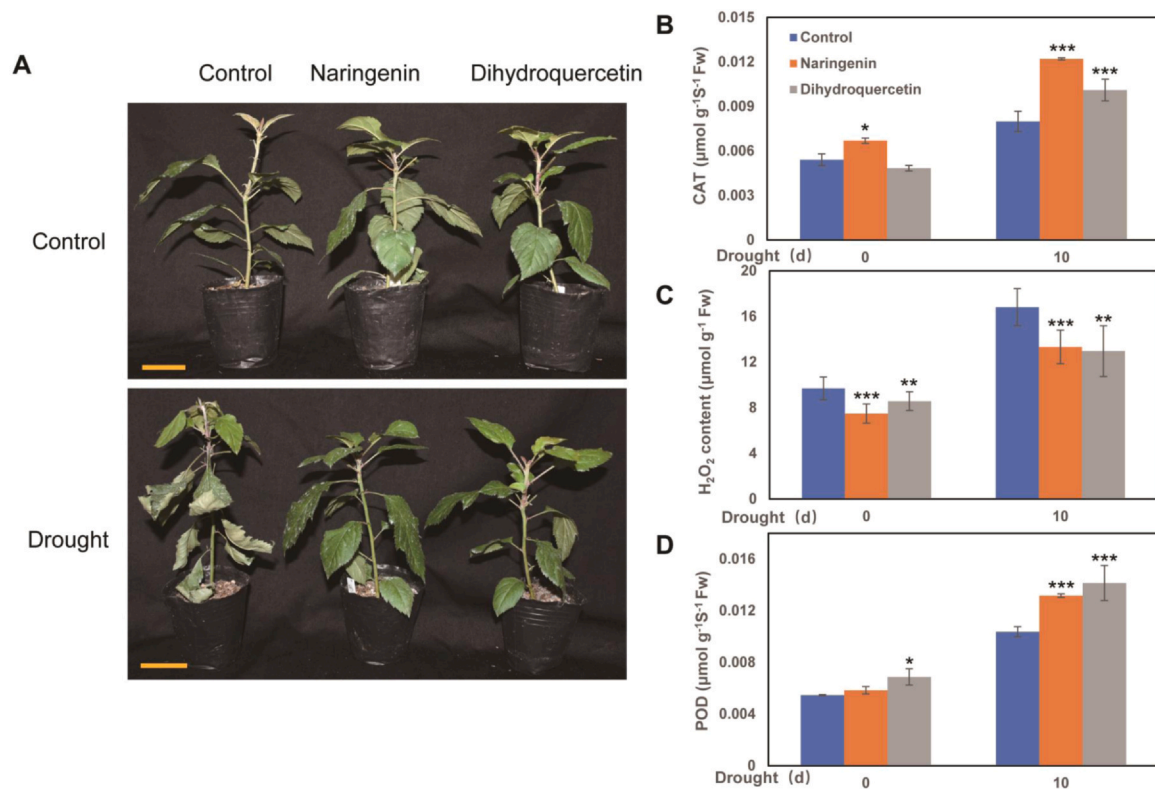


Fig. 6. The effects of naringenin-treatment and dihydroquercetin-treatment on apple ‘GL–3’ plants under drought stress. (A) Morphological characteristics of naringenin-treatment and dihydroquercetin-treatment plants with drought stress treatment. Bars, 5 cm. (B) Catalase (CAT) activity. (C) Hydrogen peroxide (H_2O_2) content. (D) Peroxidase (POD) activity. The asterisks indicate significant differences between the control GL–3 plants and naringenin-, dihydroquercetin-treatment plants based on Student’s two tailed *t*-test (*, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$). The error bars indicate standard deviations ($n = 7$).

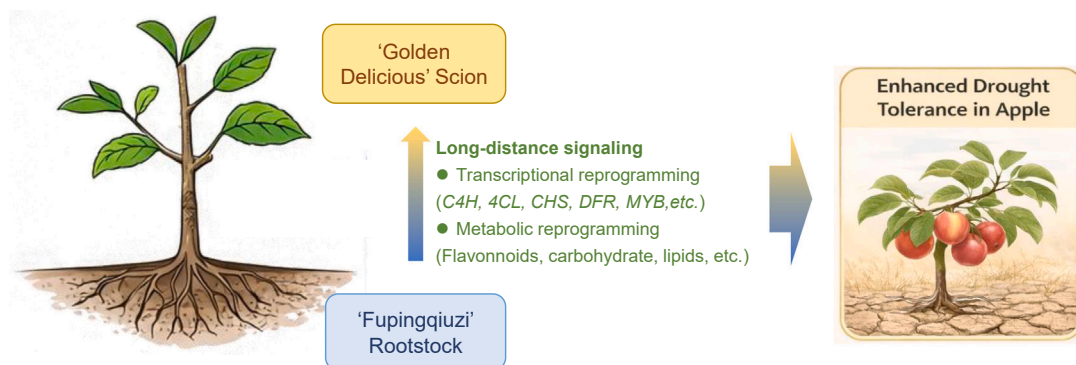


Fig. 7. Proposed model of grafting-mediated stress tolerance. Grafting onto ‘FPQZ’ enhances stress tolerance through coordinated transcriptional and metabolic reprogramming, particularly via flavonoid biosynthesis.

affecting tree architecture particularly the dwarfing effect; and promoting nutrient and mineral uptake by the scion in order to shorten the juvenile period and promote earlier flowering (Mitani et al., 2008; Silva Souza et al., 2018). An American *Vitis* species with a natural tolerance to the soil-dwelling insect pest *Phylloxera* rescued the European grape and wine industry by serving as a rootstock during the late 19th century (Ollat et al., 2016). *Citrus* rootstocks with various root architectures aid in the absorption of water and nutrients (Dumont et al., 2016; Sorgonà et al., 2007). Drought-tolerant rootstocks promote water use efficiency of apple scions under drought conditions (Liu et al., 2012; Sahin et al., 2017). Compared with a drought-sensitive grapevine rootstock, a drought-tolerant rootstock improves resistance of the scion to water stress by increasing its stomatal conductance and transpiration (Berdeja et al., 2015; Peccoux et al., 2018). Scions (Golden Delicious) grafted onto *M. prunifolia* ‘FPQZ’ showed an increased tolerance to drought, cold, heat stress, and pathogen infection (Li et al., 2022). In addition,

‘Golden Delicious’ grafted onto *M. prunifolia* ‘FPQZ’ increased expression of genes related to stress tolerance and altered DNA methylation status (Li et al., 2022). In the current study, ‘Golden Delicious’ grafted onto *M. prunifolia* ‘FPQZ’ showed higher contents of metabolites involved in biosynthesis of flavonoid, flavone and flavanol, and isoflavonoid biosynthesis, such as naringenin chalcone, naringenin, apigenin, dihydroquercetin, quercetin, phlorizin and so on. Our data indicated that increased tolerance of ‘Golden Delicious’ grafted onto *M. prunifolia* ‘FPQZ’ might be attributed to these increased metabolites. Interestingly, metabolites in carbon metabolism were also increased in ‘Golden Delicious’ grafted onto *M. prunifolia* ‘FPQZ’ like ribulose–5-phosphate, D-Glucose 6-phosphate, D-Fructose 6-phosphate, glucose–1-phosphate, D-Fructose 6-phosphate-disodium salt, indicating that other metabolism pathway should also contribute to the increased tolerance of ‘Golden Delicious’. Based on our findings, we proposed a conceptual model illustrating how grafting onto ‘FPQZ’ enhances stress tolerance

through coordinated transcriptional and metabolic reprogramming, particularly via flavonoid biosynthesis (Fig. 7).

Author contribution statement

QG designed the project. JG, JH and CN performed the experiments. JG and JH analyzed the data. JG, AK and XL wrote the manuscript. All authors read and approved the manuscript.

CRedit authorship contribution statement

Xuewei Li: Project administration, Investigation, Funding acquisition. **Junxing Guo:** Writing – original draft, Visualization, Investigation. **Jieqiang He:** Software, Investigation. **Abid Khan:** Writing – review & editing. **Chundong Niu:** Investigation, Funding acquisition. **Qingmei Guan:** Project administration, Investigation, Funding acquisition, Data curation.

Funding

This work was funded by National Key Research and Development Project (2025YFE0111900) and the National Natural Science Foundation of China (32572978, 32402520).

Data availability

All datasets generated for this study are included in the article/Supplementary Files.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Qingmei Guan reports financial support was provided by National Key Research and Development Project of China. Chundong Niu, Xuewei Li reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank the High-Performance Computing (HPC) platform of Northwest A&F University (NWAU) for providing computing resources.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jih.2026.100004](https://doi.org/10.1016/j.jih.2026.100004).

References

- Al-Harbi, A.R., Al-Omran, A.M., Alharbi, K., 2018. Grafting improves cucumber water stress tolerance in Saudi Arabia. *Saudi J. Biol. Sci.* 25, 298–304.
- Aloni, B., Cohen, R., Karni, L., Aktas, H., Edelstein, M., 2010. Hormonal signaling in rootstock–scion interactions. *Sci. Hortic.* 127, 119–126.
- Avramidou, E., Kapazoglou, A., Aravanopoulos, F.A., Xanthopoulou, A., Ganopoulos, I., Tsaballa, A., Madesis, P., Doulis, A.G., Tsaftaris, A., 2015. Global DNA methylation changes in Cucurbitaceae inter-species grafting. *Crop. Breed. Appl. Biotechnol.* 15, 112–116.
- Berdeja, M., Nicolas, P., Kappel, C., Dai, Z.W., Hilbert, G., Peccoux, A., Lafontaine, M., Ollat, N., Gomès, E., Delrot, S., 2015. Water limitation and rootstock genotype interact to alter grape berry metabolism through transcriptome reprogramming. *Hortic. Res.* 2, 15012.
- Chen, W., et al., 2014. Genome-wide association analyses provide genetic and biochemical insights into natural variation in rice metabolism. *Nat. Genet.* 46, 714–721.
- Chen, P., Yan, M., Li, L., He, J., Zhou, S., Li, Z., Niu, C., Bao, C., Zhi, F., Ma, F., Guan, Q., 2020. The apple DNA-binding one zinc-finger protein MdDof54 promotes drought resistance. *Hortic. Res.* 7, 195.
- Dong, D., Shi, Y.-N., Mou, Z.-M., Chen, S.-Y., Zhao, D.-K., 2022. Grafting: a potential method to reveal the differential accumulation mechanism of secondary metabolites. *Hortic. Res.* 9, uhac050.
- Dumont, C., Cochetel, N., Lauvergeat, V., Cookson, S.J., Ollat, N., Vivin, P., 2016. Screening root morphology in grafted grapevine using 2D digital images from rhizotrons. *Acta Hortic.* 213–220.
- Flaishman, M.A., Loginovsky, K., Golobowich, S., Lev-Yadun, S., 2008. Arabidopsis thaliana as a Model System for Graft Union Development in Homografts and Heterografts. *J. Plant. Growth Regul.* 27, 231.
- Geng, D., Shen, X., Xie, Y., Yang, Y., Bian, R., Gao, Y., Li, P., Sun, L., Feng, H., Ma, F., Guan, Q., 2020. Regulation of phenylpropanoid biosynthesis by MdMYB88 and MdMYB124 contributes to pathogen and drought resistance in apple. *Hortic. Res.* 7, 102.
- Ghanem, M.E., Albacete, A., Smigocki, A.C., Frébort, I., Pospíšilová, H., Martínez-Andújar, C., Acosta, M., Sánchez-Bravo, J., Lutts, S., Dodd, I.C., Pérez-Alfocea, F., 2011. Root-synthesized cytokinins improve shoot growth and fruit yield in salinized tomato (*Solanum lycopersicum* L.) plants. *J. Exp. Bot.* 62, 125–140.
- Gilardi, G., Baudino, M., Moizio, M., Pugliese, M., Garibaldi, A., Gullino, M.L., 2013. Integrated management of *Phytophthora capsici* on bell pepper by combining grafting and compost treatment. *Crop. Prot.* 53, 13–19.
- Haywood, V., Yu, T.-S., Huang, N.-C., Lucas, W.J., 2005. Phloem long-distance trafficking of GIBBERELLIC ACID-INSENSITIVE RNA regulates leaf development: phloem delivery of RNA regulates leaf development. *Plant. J.* 42, 49–68.
- Hou, Q., Ufer, G., Bartels, D., 2016. Lipid signalling in plant responses to abiotic stress. *Plant. Cell. Environ.* 39, 1029–1048.
- Huang, L., Li, M., Zhou, K., Sun, T., Hu, L., Li, C., Ma, F., 2018. Uptake and metabolism of ammonium and nitrate in response to drought stress in *Malus prunifolia*. *Plant. Physiol. Biochem.* 127, 185–193.
- Huang, Y., Xu, Y., Jiang, X., Yu, H., Jia, H., Tan, C., Hu, G., Hu, Y., Rao, M.J., Deng, X., Xu, Q., 2021. Genome of a citrus rootstock and global DNA demethylation caused by heterografting. *Hortic. Res.* 8, 69.
- Ikinci, A., Bolat, I., Ercisli, S., Kodad, O., 2014. Influence of rootstocks on growth, yield, fruit quality and leaf mineral element contents of pear cv. ‘Santa Maria’ in semi-arid conditions. *Biol. Res.* 47, 71.
- Kanehisa, M., Goto, S., 2000. KEGG: Kyoto Encyclopedia of Genes and Genomes. *Nucleic Acids Res.* 28 (1), 27–30.
- Koepke, T., Dhirga, A., 2013. Rootstock scion somatogenetic interactions in perennial composite plants. *Plant. Cell. Rep.* 32, 1321–1337.
- Koyama, K., Numata, M., Nakajima, I., Goto-Yamamoto, N., Matsumura, H., Tanaka, N., 2014. Functional characterization of a new grapevine MYB transcription factor and regulation of proanthocyanidin biosynthesis in grapes. *J. Exp. Bot.* 65, 4433–4449.
- Lardizabal, R.D., Thompson, P.G., 1990. Growth regulators combined with grafting increase flower number and seed production in Sweet Potato. *HortScience* 25, 79–81.
- Li, Z., et al., 2022. Chromosome-scale reference genome provides insights into the genetic origin and grafting-mediated stress tolerance of *Malus prunifolia*. *Plant. Biotechnol. J.* 20, 1015–1017.
- Li, M., Liu, J., Zhou, Y., Zhou, S., Zhang, S., Tong, H., Zhao, A., 2020. Transcriptome and metabolome profiling unveiled mechanisms of tea (*Camellia sinensis*) quality improvement by moderate drought on pre-harvest shoots. *Phytochemistry* 180, 112515.
- Li, Y., Sun, H., Li, J., Qin, S., Yang, W., Ma, X., Qiao, X., Yang, B., 2021. Effects of Genetic Background and Altitude on Sugars, Malic Acid and Ascorbic Acid in Fruits of Wild and Cultivated Apples (*Malus* sp.). *Foods* 10, 2950.
- Liang, L., Gao, C., Luo, M., Wang, W., Zhao, C., Zu, Y., Efferth, T., Fu, Y., 2013. Dihydroquercetin (DHQ) Induced HO-1 and NQO1 Expression against Oxidative Stress through the Nrf2-Dependent Antioxidant Pathway. *J. Agric. Food Chem.* 61, 2755–2761.
- Liu, B.H., Cheng, L., Liang, D., Zou, Y.J., Ma, F.W., 2012. Growth, gas exchange, water-use efficiency, and carbon isotope composition of ‘Gala’ apple trees grafted onto 9 wild Chinese rootstocks in response to drought stress. *Photosynthetica* 50, 401–410.
- Liu, W., Xiang, C., Li, X., Wang, T., Lu, X., Liu, Z., Gao, L., Zhang, W., 2020b. Identification of Long-Distance Transmissible mRNA between Scion and Rootstock in Cucurbit Seedling Heterografts. *Int. J. Mol. Sci.* 21, 5253.
- Liu, L., Zhang, Y., Yu, H., 2020a. Florigen trafficking integrates photoperiod and temperature signals in Arabidopsis. *J. Integr. Plant Biol.* 62, 1385–1398.
- Martin, A., Adam, H., Díaz-Mendoza, M., Żurczak, M., González-Schain, N.D., Suárez-López, P., 2009. Graft-transmissible induction of potato tuberization by the microRNA miR172. *Development* 136, 2873–2881.
- Mekawy, A.M.M., Abdelaziz, M.N., Ueda, A., 2018. Apigenin pretreatment enhances growth and salinity tolerance of rice seedlings. *Plant. Physiol. Biochem.* 130, 94–104.
- Mitani, N., Matsumoto, R., Yoshioka, T., Kuniga, T., 2008. Citrus hybrid seedlings reduce initial time to flower when grafted onto shikukusha rootstock. *Sci. Hortic.* 116, 452–455.
- Moore, R., 1982. Graft Formation in *Kalanchoe blossfeldiana*. *J. Exp. Bot.* 33, 533–540.
- Mudge, K., Janick, J., Scofield, S., Goldschmidt, E.E., 2009. In: *Horticultural Reviews*, Janick, J. (Eds.), A History of Grafting. John Wiley & Sons, Inc, Hoboken, NJ, USA, pp. 437–493.
- Nakabayashi, R., et al., 2014. Enhancement of oxidative and drought tolerance in Arabidopsis by overaccumulation of antioxidant flavonoids. *Plant. J.* 77, 367–379.
- North, M.S., Cook, N.C., 2008. Effect of six rootstocks on ‘forelle’ pear tree growth, production, fruit quality and leaf mineral content. *Acta Hortic.* 97–103.
- Okazaki, Y., Saito, K., 2014. Roles of lipids as signaling molecules and mitigators during stress response in plants. *Plant. J.* 79, 584–596.
- Ollat, N., Bordenave, L., Tandonnet, J.P., Boursiquot, J.M., Marguerit, E., 2016. Grapevine rootstocks: origins and perspectives. *Acta Hortic.* 11–22.

- Osorio, S., Carneiro, R.T., Lytovchenko, A., McQuinn, R., Sørensen, I., Vallarino, J.G., Giovannoni, J.J., Fernie, A.R., Rose, J.K.C., 2020. Genetic and metabolic effects of ripening mutations and vine detachment on tomato fruit quality. *Plant. Biotechnol. J.* 18, 106–118.
- Ozfidan-Konakci, C., Yildiztugay, E., Alp, F.N., Kucukoduk, M., Turkan, I., 2020. Naringenin induces tolerance to salt/osmotic stress through the regulation of nitrogen metabolism, cellular redox and ROS scavenging capacity in bean plants. *Plant. Physiol. Biochem.* 157, 264–275.
- Pant, B.D., Buhtz, A., Kehr, J., Scheible, W.-R., 2008. MicroRNA399 is a long-distance signal for the regulation of plant phosphate homeostasis. *Plant. J.* 53, 731–738.
- Peccoux, A., Loveys, B., Zhu, J., Gambetta, G.A., Delrot, S., Vivin, P., Schultz, H.R., Ollat, N., Dai, Z., 2018. Dissecting the rootstock control of scion transpiration using model-assisted analyses in grapevine. *Tree Physiol.* 38, 1026–1040.
- Pina, A., Errea, P., 2005. A review of new advances in mechanism of graft compatibility-incompatibility. *Sci. Hortic.* 106, 1–11.
- Qi, D., Li, J., Qiao, X., Lu, M., Chen, W., Miao, A., Guo, W., Ma, C., 2019. Non-targeted metabolomic analysis based on ultra-high-performance liquid chromatography quadrupole time-of-flight tandem mass spectrometry reveals the effects of grafting on non-volatile metabolites in fresh tea leaves (*Camellia sinensis* L.). *J. Agric. Food Chem.* 67, 6672–6682.
- Reeves, G., Tripathi, A., Singh, P., Jones, M.R.W., Nanda, A.K., Musseau, C., Craze, M., Bowden, S., Walker, J.F., Bentley, A.R., Melnyk, C.W., Hibberd, J.M., 2022. Monocotyledonous plants graft at the embryonic root-shoot interface. *Nature* 602, 280–286.
- Sahin, O., Gunes, A., Taskin, M.B., Inal, A., 2017. Investigation of responses of some apple (*Mallus x domestica* Borkh.) cultivars grafted on MM106 and M9 rootstocks to lime-induced chlorosis and oxidative stress. *Sci. Hortic.* 219, 79–89.
- Shahid, M.A., Balal, R.M., Khan, N., Simón-Grao, S., Alfósea-Simón, M., Cámara-Zapata, J.M., Mattson, N.S., García-Sánchez, F., 2019. Rootstocks influence the salt tolerance of Kinnow mandarin trees by altering the antioxidant defense system, osmolyte concentration, and toxic ion accumulation. *Sci. Hortic.* 250, 1–11.
- Shi, X., Wang, X., Cheng, F., Cao, H., Liang, H., Lu, J., Kong, Q., Bie, Z., 2019. iTRAQ-based quantitative proteomics analysis of cold stress-induced mechanisms in grafted watermelon seedlings. *J. Proteom.* 192, 311–320.
- Sidorova, T., Miroshnichenko, D., Kirov, I., Pushin, A., Dolgov, S., 2021. Effect of grafting on viral resistance of non-transgenic plum scion combined with transgenic PPV-Resistant Rootstock. *Front. Plant. Sci.* 12, 56.
- Silva Souza, L., Diniz, R.P., Neves, R., de, J., Alves, A.A.C., Oliveira, E.J. de, 2018. Grafting as a strategy to increase flowering of cassava. *Sci. Hortic.* 240, 544–551.
- Sokolov, V.V., Savell'ev, N.I., Goncharov, N.P., 2015. I. V. Michurin's work on expansion of the plant horticulture assortment and improvement of food quality. *Proc. Latv. Acad. Sci. Sect. B. Nat. Exact. Appl. Sci.* 69, 190–197.
- Sorgonà, A., Abenavoli, M.R., Gringeri, P.G., Lupini, A., Cacco, G., 2007. Root architecture plasticity of citrus rootstocks in response to nitrate availability. *J. Plant. Nutr.* 30, 1921–1932.
- Stincione, A., et al., 2015. The return of metabolism: biochemistry and physiology of the pentose phosphate pathway. *Biol. Rev.* 90, 927–963.
- Sun, H., Li, Q., Mao, L.-Z., Yuan, Q.-L., Huang, Y., Chen, M., Fu, C.-F., Zhao, X.-H., Li, Z.-Y., Dai, Y.-H., Zou, X.-X., Ou, L.-J., 2021. Investigating the molecular mechanisms of pepper fruit tolerance to storage via transcriptomics and metabolomics. *Horticulturae* 7, 242.
- Tedesco, S., Erban, A., Gupta, S., Kopka, J., Fevèreiro, P., Kragler, F., Pina, A., 2021. The impact of metabolic scion-rootstock interactions in different grapevine tissues and phloem exudates. *Metabolites* 11, 349.
- Thieme, C.J., Rojas-Triana, M., Stecyk, E., Schudoma, C., Zhang, W., Yang, L., Miñambres, M., Walther, D., Schulze, W.X., Paz-Ares, J., Scheible, W.-R., Kragler, F., 2015. Endogenous Arabidopsis messenger RNAs transported to distant tissues. *Nat. Plants* 1, 15025.
- Tietel, Z., Lewinsohn, E., Fallik, E., Porat, R., 2011. Elucidating the Roles of Ethanol Fermentation Metabolism in Causing Off-Flavors in Mandarins. *J. Agric. Food Chem.* 59, 11779–11785.
- Tsipouridis, C., Thomidis, T., 2005. Effect of 14 peach rootstocks on the yield, fruit quality, mortality, girth expansion and resistance to frost damages of May Crest peach variety and their susceptibility on *Phytophthora citrophthora*. *Sci. Hortic.* 103, 421–428.
- Turhan, A., Ozmen, N., Kuscü, H., Serbeci, M.S., Seniz, V., 2012. Influence of rootstocks on yield and fruit characteristics and quality of watermelon. *Hortic. Environ. Biotechnol.* 53, 336–341.
- Upchurch, R.G., 2008. Fatty acid unsaturation, mobilization, and regulation in the response of plants to stress. *Biotechnol. Lett.* 30, 967–977.
- Venema, J.H., Dijk, B.E., Bax, J.M., van Hasselt, P.R., Elzenga, J.T.M., 2008. Grafting tomato (*Solanum lycopersicum*) onto the rootstock of a high-altitude accession of *Solanum habrochaites* improves suboptimal-temperature tolerance. *Environ. Exp. Bot.* 63, 359–367.
- Viehweiger, K., Dordschbal, B., Roos, W., 2002. Elicitor-activated phospholipase A₂ generates lysophosphatidylcholines that mobilize the vacuolar H⁺ pool for pH signaling via the activation of Na⁺-dependent proton fluxes. *Plant. Cell.* 14, 1509–1525.
- Wan, H., Yang, F., Zhuang, X., Cao, Y., He, J., Li, H., Qin, S., Lyu, D., 2021. Malus rootstocks affect copper accumulation and tolerance in trees by regulating copper mobility, physiological responses, and gene expression patterns. *Environ. Pollut.* 287, 117610.
- Wang, Y., Kollmann, R., 1996. Vascular differentiation in the graft union of in-vitro grafts with different compatibility. — structural and functional aspects. *J. Plant. Physiol.* 147, 521–533.
- Wang, S., Liang, D., Li, C., Hao, Y., Ma, F., Shu, H., 2012. Influence of drought stress on the cellular ultrastructure and antioxidant system in leaves of drought-tolerant and drought-sensitive apple rootstocks. *Plant. Physiol. Biochem.* 51, 81–89.
- Wang, Y., Wang, L., Xing, N., Wu, X., Wu, X., Wang, B., Lu, Z., Xu, P., Tao, Y., Li, G., Wang, Y., 2020. A universal pipeline for mobile mRNA detection and insights into heterografting advantages under chilling stress. *Hortic. Res.* 7, 13.
- Warschafsky, E.J., Klein, L.L., Frank, M.H., Chitwood, D.H., Londo, J.P., von Wettberg, E.J.B., Miller, A.J., 2016. Rootstocks: diversity, domestication, and impacts on shoot phenotypes. *Trends Plant. Sci.* 21, 418–437.
- Wei, Y., Zhou, D., Peng, J., Pan, L., Tu, K., 2017. Hot air treatment induces disease resistance through activating the phenylpropanoid metabolism in cherry tomato fruit. *J. Agric. Food Chem.* 65, 8003–8010.
- Xu, J., Yan, J., Li, W., Wang, Q., Wang, C., Guo, J., Geng, D., Guan, Q., Ma, F., 2020. Integrative analyses of widely targeted metabolic profiling and transcriptome data reveals molecular insight into metabolomic variations during apple (*Malus domestica*) fruit development and ripening. *Int. J. Mol. Sci.* 21, 4797.
- Yaginuma, S., Shiraishi, T., Ohya, H., Igarashi, K., 2002. Polyphenol increases in safflower and cucumber seedlings exposed to strong visible light with limited water. *Biosci. Biotechnol. Biochem.* 66, 65–72.
- Yang, Y., Mao, L., Jittayasothorn, Y., Kang, Y., Jiao, C., Fei, Z., Zhong, G.-Y., 2015. Messenger RNA exchange between scions and rootstocks in grafted grapevines. *BMC Plant. Biol.* 15, 251.
- Yang, C., Yang, H., Xu, Q., Wang, Y., Sang, Z., Yuan, H., 2020. Comparative metabolomics analysis of the response to cold stress of resistant and susceptible Tibetan hullless barley (*Hordeum distichon*). *Phytochemistry* 174, 112346.
- Yoon, J., Cho, L.-H., Tun, W., Jeon, J.-S., An, G., 2021. Sucrose signaling in higher plants. *Plant. Sci.* 302, 110703.
- Zheng, Y., Wang, P., Chen, X., Sun, Y., Yue, C., Ye, N., 2019. Transcriptome and metabolite profiling reveal novel insights into volatile heterosis in the tea plant (*Camellia sinensis*). *Molecules* 24, 3380.